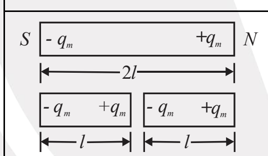
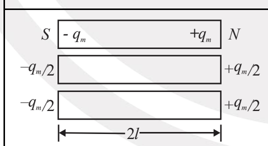
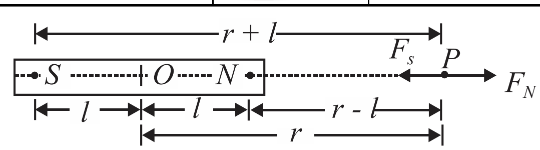
CHAPTER 5

Magnetism and Matter

<table>
<tr><th>Sr. No.</th><th>Concept</th><th>Formulas</th><th>Other information</th></tr>
<tr><td>1.</td><td>Coulomb's law of magnetism and dipole moment of a magnet</td><td>Magnetic dipole moment, $m = q_m \times 2l$

Coulomb's law, $F = \frac{\mu_0}{4\pi}\frac{q_{m_1} q_{m_2}}{r^2}$

Where, μ_0 is the permeability of free space and its value is $4\pi \times 10^{-7}$ henry/meter or TmA^{-1}.</td><td>q_{m_1}, q_{m_2} = the pole strengths of two magnetic poles.
F = the force of attraction or repulsion
$2l$ = the magnetic length.</td></tr>
<tr><td>2.</td><td>Pole strength and magnetic dipole moment in special cases</td><td>
<table>
<tr><th>Figure</th><th>Effect on pole strength</th><th>Formula for new magnetic dipole moment</th></tr>
<tr><td>
</td><td>Remains unchanged</td><td>$M' = q_m.\frac{2l}{2} = \frac{M}{2}$
(become half)</td></tr>
<tr><td>
</td><td>Pole strength of each piece becomes half</td><td>$M' = \left(\frac{q_m}{2}\right)2l = \frac{M}{2}$
(becomes half)</td></tr>
</table>
</td><td>$(+q_m, -q_m)$ = strength of N-pole and S-pole.
M' = final magnetic dipole moment of each part.
$M = q_m\, 2l$ (Initial magnetic dipole moment)</td></tr>
<tr><td>3.</td><td>Magnetic field of a bar magnet at an axial point (B_{axial})</td><td>

(i) $B_{axial} = \frac{\mu_0}{4\pi}.\frac{2mr}{\left(r^2 - l^2\right)^2}$

Where r is the distance of the point from the centre of the magnet.
(ii) For a short magnet, $l << r$,
$$B_{axial} = \frac{\mu_0}{4\pi}.\frac{2m}{r^3}$$</td><td>$2l$ = length of bar magnet.
m = magnetic dipole moment of the magnet.</td></tr>
</table>

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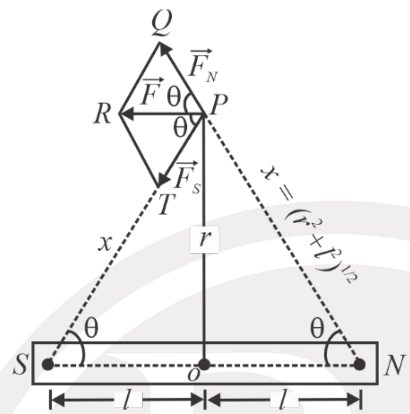
4.	**Magnetic field of a bar magnet at an equatorial point (B_{equa}).**	(i) $B_{equa} = \frac{\mu_0}{4\pi} \cdot \frac{m}{\left(r^2 + l^2\right)^{3/2}}$ (ii) For short magnet, $l << r$, $B_{equa} = \frac{\mu_0}{4\pi} \cdot \frac{m}{r^3}$  	B_{equa} = magnetic field of a bar magnet at an equatorial point. r = distance of point P from midpoint of magnet.
5.	**Torque on a magnet in a magnetic field**	Torque acting on the magnet is; $\tau = mB\sin\theta$ In vector notation, $\vec{\tau} = \vec{m} \times \vec{B}$ Torque is maximum: If $\theta = 90°$, then $\tau = mB$	τ = torque acting on the magnet. θ = angle between magnetic dipole moment m and magnetic field B.
6.	**Potential energy of a magnetic dipole in a magnetic field**	$W = \Delta U = -mB\,(\cos\theta_2 - \cos\theta_1)$ P.E. is zero when $\vec{m} \perp \vec{B}$. Hence P.E. of the dipole in any orientation θ is: $U = -mB\cos\theta = -\vec{m}.\vec{B}$	W = work done when a magnetic dipole is rotated in a magnetic field θ_1 = initial position θ_2 = final position
7.	**Current loop as a magnetic dipole**	Magnetic dipole moment $m = IA$ In vector notation, $\vec{m} = I\,\vec{A}$	A = area of current loop. I = current flowing in the loop.
8.	**Gauss's law in magnetism**	Mathematically, $\phi_B = \oint_S \vec{B}.d\vec{S} = 0$	B = magnetic field. dS = small surface area of the loop.

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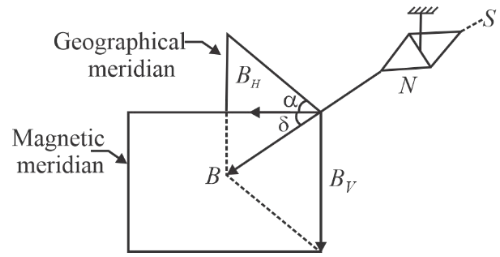
9.	**Horizontal and vertical component of earth's magnetic field.**	 $B_H = B\cos\delta$ and $B_V = B\sin\delta$ $\therefore \dfrac{B_V}{B_H} = \dfrac{B\sin\delta}{B\cos\delta}$ or $\boxed{\dfrac{B_V}{B_H} = \tan\delta}$ Also, $B_H^2 + B_V^2 = B^2(\cos^2\delta + \sin^2\delta) = B^2$ or, $\boxed{B = \sqrt{B_H^2 + B_V^2}}$ At the magnetic equator, $\delta = 0°$, $B_H = B\cos 0° = B$ At the magnetic poles, $\delta = 90°$, $B_H = B\cos 90° = 0$	ϕ_B = net magnetic flux through closed surface S. B_H = horizontal component of earth's magnetic field. B_V = vertical component of earth's magnetic field. B = resultant magnetic field α = magnetic declination δ = angle of dip
10.	**Magnetising field intensity and magnetic field**	Magnetising field intensity (H): $H = nI$ $B_0 = \mu_0 nI = \mu_0 H$ $H = \dfrac{B_0}{\mu_0}$	μ_0 = permeability of free space B_0 = magnetic field n = number of turns per unit length. I = amount of current flowing through the material.
11.	**Magnetic properties of material**	Intensity of magnetisation (M): $\vec{M} = \dfrac{\vec{m}_{net}}{V}$ or $M = \dfrac{m_{net}}{V}$ It is a vector quantity. Magnetic induction (B): $B = \mu_0(H + M)$ Magnetic permeability (μ): $\mu = \dfrac{B}{H}$ Magnetic susceptibility (χ): $\chi_m = \dfrac{M}{H}$ It can be shown that $\mu = \mu_0(1 + \chi_m)$ and $\mu_r = 1 + \chi_m$	μ_r = relative magnetic permeability H = magnetising field intensity χ_m = magnetic susceptibility of a medium. V = volume of the magnetic substance m_{net} = net magnetic dipole moment of substance

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12.	**Curie's law and Curie temperature**	For paramagnetic substances: $\chi_m \propto \frac{1}{T}$ T is the absolute temperature of substance or $\chi_m = \frac{C}{T}$ For ferromagnetic substances $\chi_m = \frac{C'}{T - T_C} \quad (T > T_C) \quad [C' = \text{curie constant}]$	T_C = the temperature above which a ferromagnetic substance becomes paramagnetic. C = constant here. χ_m = magnetic susceptibility of a substance.
13.	**Time period of a bar magnet in earth's horizontal magnetic field**	Period of vibration of a bar magnet: $T = 2\pi\sqrt{\frac{I}{mB_H}}$ Tangent law: the magnet comes to rest after making an angle θ with the direction of B_H such that, $B = B_H \tan\theta$	I = moment of inertia B_H = earth's horizontal component T = time period of a oscillating bar magnet. B = resultant magnetic field.